

Homework**Polona Oblak**

For $p, q \in \mathbb{R}_2[x]$ let us define

$$g(p, q) = \int_0^1 p(x)q(x) dx.$$

- (1) Prove g is inner product on $\mathbb{R}_2[x]$.
- (2) Use Gramm-Schmidt algorithm to find an orthonormal basis $\mathcal{B}$ for $\mathbb{R}_2[x]$ with respect to g , such that $1 \in \mathcal{B}$.
- (3) (not graded) Verify that $\alpha(p) = \int_0^2 p(x) dx$ is a linear form on $\mathbb{R}_2[x]$.
- (4) Find a polynomial $r \in \mathbb{R}_2[x]$ such that $\alpha(p) = g(p, r)$ for all $p \in \mathbb{R}_2[x]$.

The solution of the problem needs to be submitted to the appropriate mailbox on Ucilnica until a day before the exam, but preferably in a week. The solutions should be written by hand (on paper and scanned or on a tablet) and should contain clear and well-explained proofs, procedures, explanations, etc. At the end, add a short statement confirming that the submitted solution is your own work. Also list the names of people you discussed the assignment with (if any) and clearly indicate which parts of the solution were developed by you and which parts were assisted by AI tools (if any).